Status	Finished
Started	Sunday, 11 October 2026, 1:07 AM
Completed	Sunday, 11 October 2026, 1:13 AM
Duration	6 mins 5 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Linear Search
c. Merge Sort
d. Binary Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting is rarely used in programming
c. Sorting makes it harder to search for items
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. All of the above
- c. Selection
- d. Duplicates

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Inserting data into a list
- c. Searching for data in a list
- d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **5**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Distribution
- b. Selection
- c. Extraction
- d. Insertion

The correct answer is:
Insertion

Question **6**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Sorting
- c. Searching
- d. Merging

The correct answer is:
Complexity

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It requires external libraries
- c. It sorts data out of the box efficiently
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Rearranging data without sorting
- c. Sorting data in a single pass
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Merge Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It increases the number of comparisons needed
- b. It makes the search operation slower
- c. It has no effect on the search operation
- d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **11**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Merging
- b. Sorting
- c. Searching
- d. Rearranging

The correct answer is: Sorting

Question **12**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. sort()
- c. sorted()
- d. order()

The correct answer is: sort()

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sorted()
- c. sort()
- d. order()

The correct answer is: sorted()

Question **15**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It makes the analysis slower
c. It allows for quicker identification of duplicates
d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates